

SN1 & SN2 (Cambridge (CIE) A Level Chemistry): Revision Note

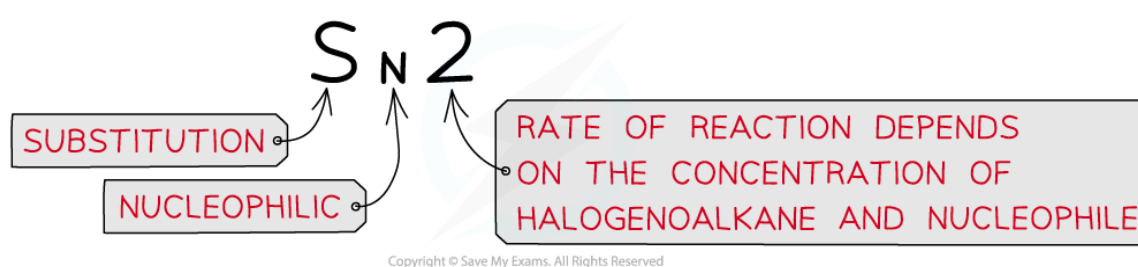
Halogenoalkanes: SN1 & SN2 Mechanisms

- In **nucleophilic substitution** reactions involving halogenoalkanes, the halogen atom is replaced by a nucleophile
- These reactions can occur in two different ways (known as **S_N2** and **S_N1** reactions) depending on the structure of the halogenoalkane involved

S_N2 reactions

- In **primary halogenoalkanes**, the carbon that is attached to the halogen is bonded to **one alkyl group**
- These halogenoalkanes undergo nucleophilic substitution by an **S_N2** mechanism
 - 'S' stands for 'substitution'
 - 'N' stands for 'nucleophilic'
 - '2' means that the rate of the reaction (which is determined by the slowest step of the reaction) depends on the concentration of both the halogenoalkane and the nucleophile ions

Defining an S_N2 mechanism

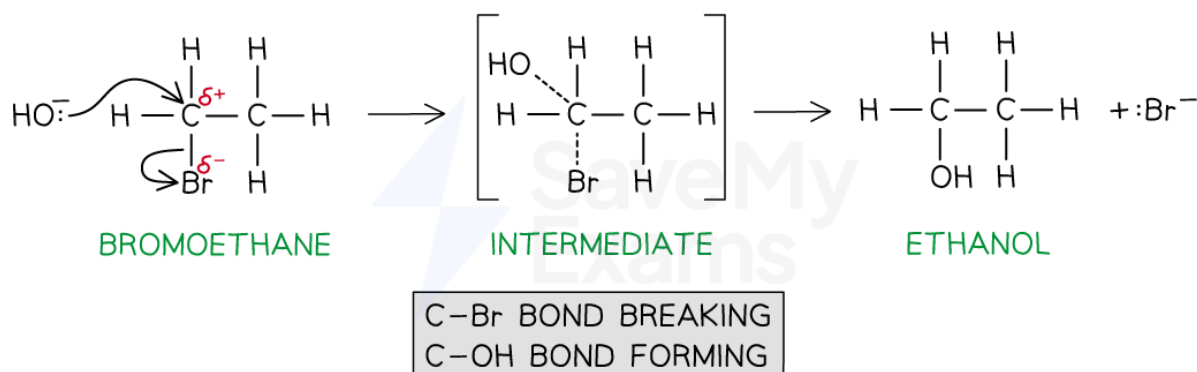


Each term in the S_N2 expression has a specific meaning

- The S_N2 mechanism is a **one-step** reaction
 - The nucleophile donates a pair of electrons to the δ⁺ carbon atom to form a new bond
 - At the same time, the C-X bond is breaking and the halogen (X) takes both electrons in the bond (**heterolytic fission**)
 - The halogen leaves the halogenoalkane as an X⁻ ion

- For example, the nucleophilic substitution of bromoethane by hydroxide ions to form ethanol

The nucleophilic substitution of bromoethane by hydroxide ions



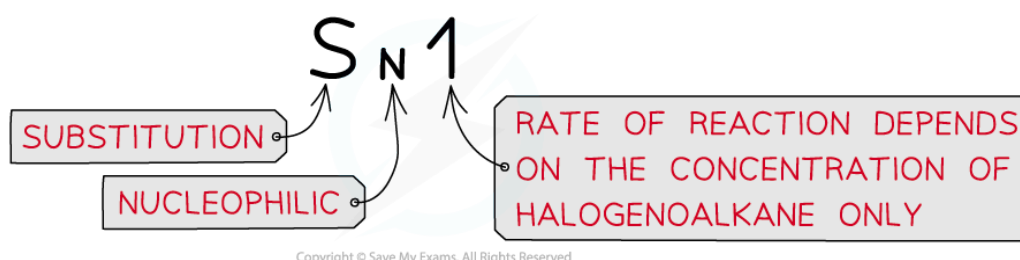
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In this mechanism, the bromoethane is a primary halogenoalkane

S_N1 reactions

- In **tertiary halogenoalkanes**, the carbon that is attached to the halogen is bonded to **three alkyl groups**
- These halogenoalkanes undergo nucleophilic substitution by an **S_N1** mechanism
 - 'S' stands for 'substitution'
 - 'N' stands for 'nucleophilic'
 - '1' means that the rate of the reaction (which is determined by the slowest step of the reaction) depends on the concentration of only one reagent, the **halogenoalkane**

Defining an S_N1 mechanism



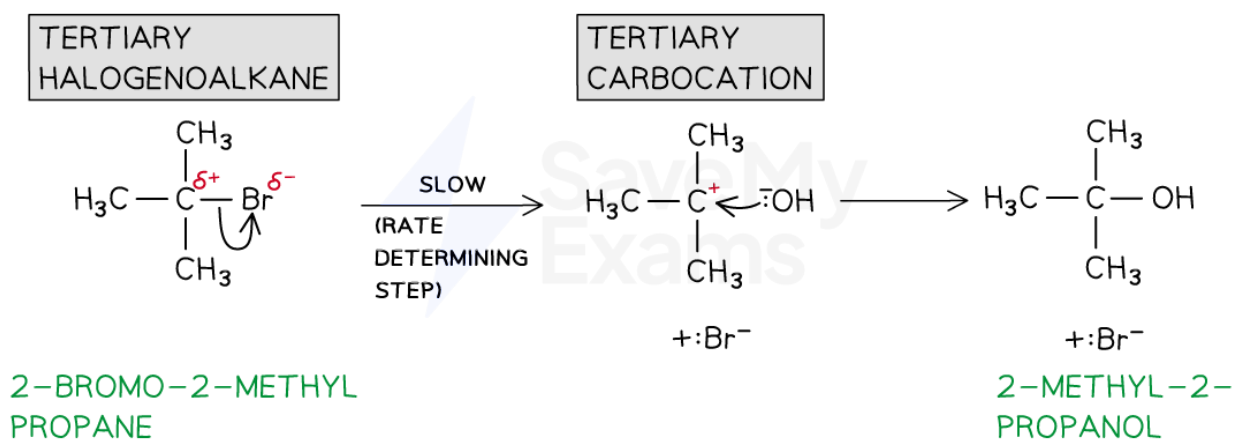
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Each term in the S_N1 expression has a specific meaning

- The S_N1 mechanism is a two-step reaction

- In the first step, the C-X bond breaks **heterolytically** and the halogen leaves the halogenoalkane as an X^- ion (this is the **slow** and **rate-determining step**)
- This forms a **tertiary carbocation (which is a tertiary carbon atom with a positive charge)**
- In the second step, the tertiary carbocation is attacked by the **nucleophile**
- For example, the nucleophilic substitution of 2-bromo-2-methylpropane by hydroxide ions to form 2-methyl-2-propanol

The nucleophilic substitution of 2-bromo-2-methylpropane by hydroxide ions

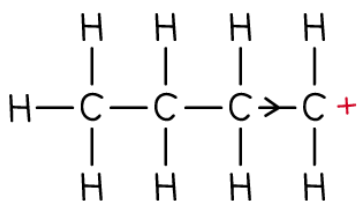


In this mechanism, the 2-bromo-2-methylpropane is a tertiary halogenoalkane

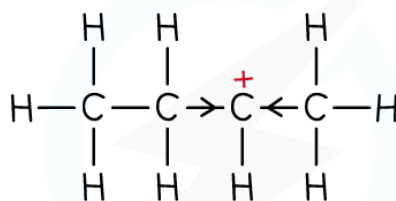
Carbocations

- In the S_N1 mechanism, a **tertiary carbocation** is formed
- This is not the case for S_N2 mechanisms as a **primary carbocation** would have been formed which is much **less** stable than tertiary carbocations
- This has to do with the **positive inductive effect** of the alkyl groups attached to the carbon which is bonded to the halogen atom
 - The alkyl groups push electron density towards the positively charged carbon, reducing the **charge density**
 - In tertiary carbocations, there are three alkyl groups stabilising the carbocation whereas in primary carbocations there is only one alkyl group
 - This is why tertiary carbocations are much more stable than primary ones

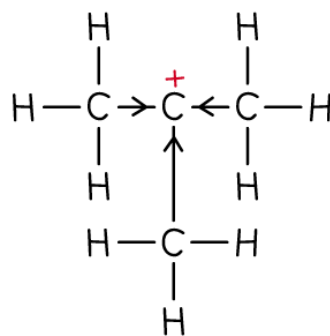
Stability of primary, secondary and tertiary carbocations



PRIMARY CARBOCATION
(LEAST STABLE)



SECONDARY
CARBOCATION



TERTIARY CARBOCATION
(MOST STABLE)

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The carbocations become more stable moving from primary to secondary to tertiary

- **Secondary halogenoalkanes** undergo a **mixture** of both S_N1 and S_N2 reactions depending on their structure